BASICS OF FIM · WI001287

Statistical Learning

3: OLS Regressions

Univ.-Prof. Dr.-Ing. habil. Niklas Kühl
Lars Böcking

Chair for Information Systems and
Human-centric Artificial Intelligence
University of Bayreuth
TUM School of Management

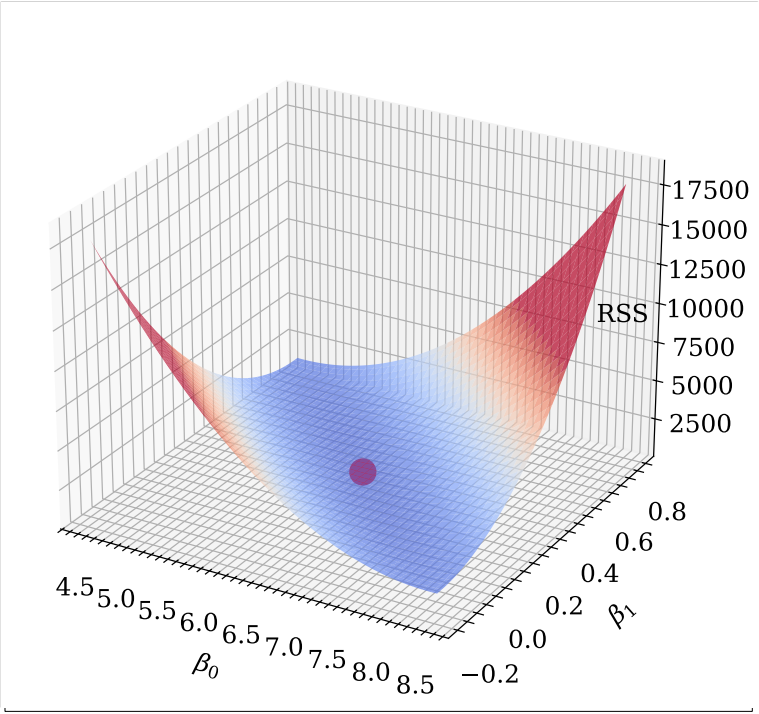


Figure 3.3: residual sum of squares surface

Agenda

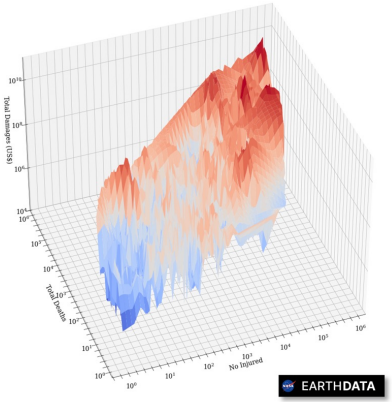
- Simple linear regression
- Multiple linear regression
- Extensions to linear regression

Agenda

- Simple linear regression
  - Estimating the Coefficients
  - Assessing the Accuracy of the Coefficient Estimates
  - Assessing the Accuracy of the Model
- Multiple linear regression
- Extensions to linear regression

Quick rewind: what we already know about learning $f(X)$.

Why have a function from X to Y in the first place?



$$\textit{Total Damage (US\$)} \approx f(\textit{Homeless}, \textit{Total Death}, \textit{Injured})$$

$$\hat{Y} \approx \hat{f}(X)$$

- With a good $\hat{f}$ we can make predictions of Y at new points $X = x$
- We can understand which components of $X = (X_1, X_2, \ldots, X_p)$ are important in explaining Y

Statistical Learning (Basics of FIM) | Prof. Dr. Niklas Kühl 10

We bet the world is a line plus noise.

Linear regression is a simple approach to supervised learning. We believe about the world that: $Y = f(X) + \varepsilon$. In the case of linear regression, we assume that the dependence of Y on X is linear:

y-axis intercept *error term*

$$Y = \beta_0 + \beta_1 \times X + \varepsilon$$

slope

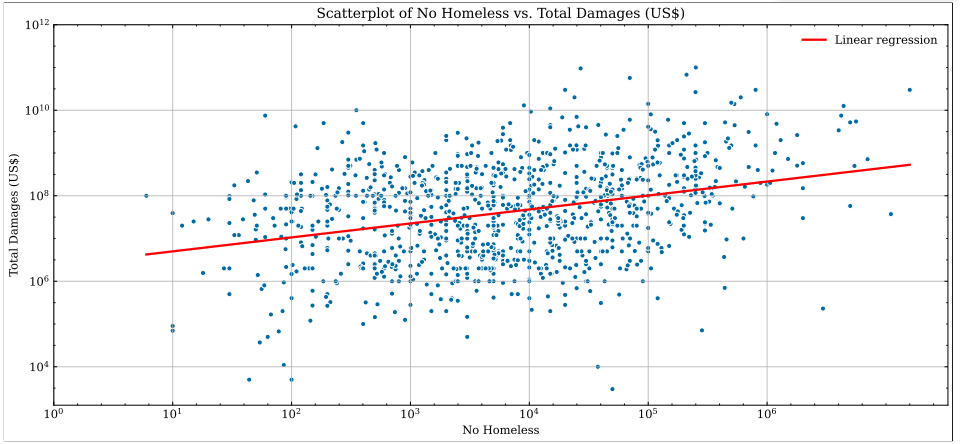


Figure 3.1: line plus noise

… but we cannot observe this entity
… instead we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^{N}$
… from which we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

True regression functions are never linear!

Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

We pick the line that minimizes squared prediction errors.

$$Y = \beta_0 + \beta_1 \times X + \varepsilon$$

β_0 and β_1 are unknown constants that we want to estimate ($\hat{\beta}_0$ and $\hat{\beta}_1$ respectively).

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$min \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

$$= min \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

… such that we minimize the error of our estimations $\hat{y}$

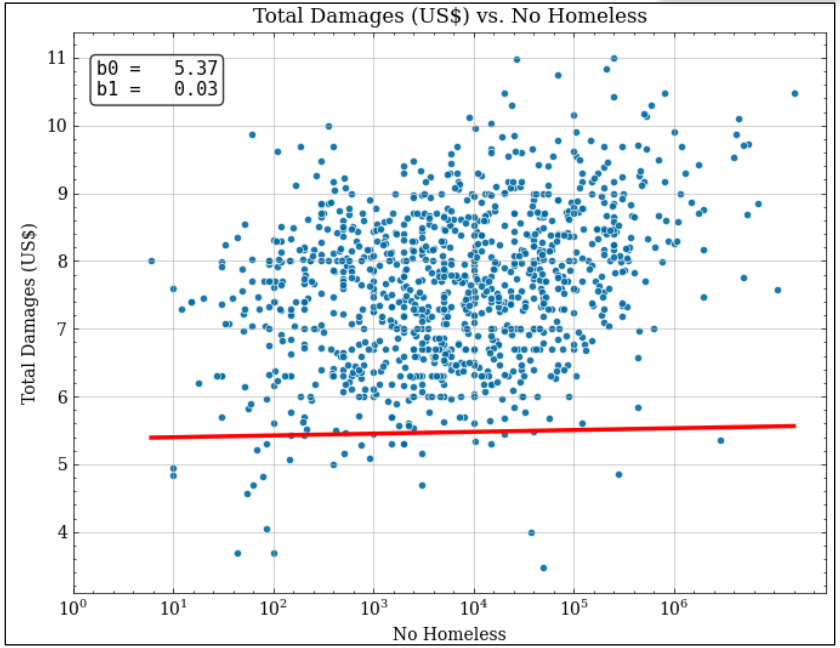


Figure 3.2: ols fit

For different combinations of our parameter estimates we can compare the residual sum of squares.

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= min \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = min \sum_{i=1}^{N} {e_i}^2$$

Residual sum of squares (RSS)

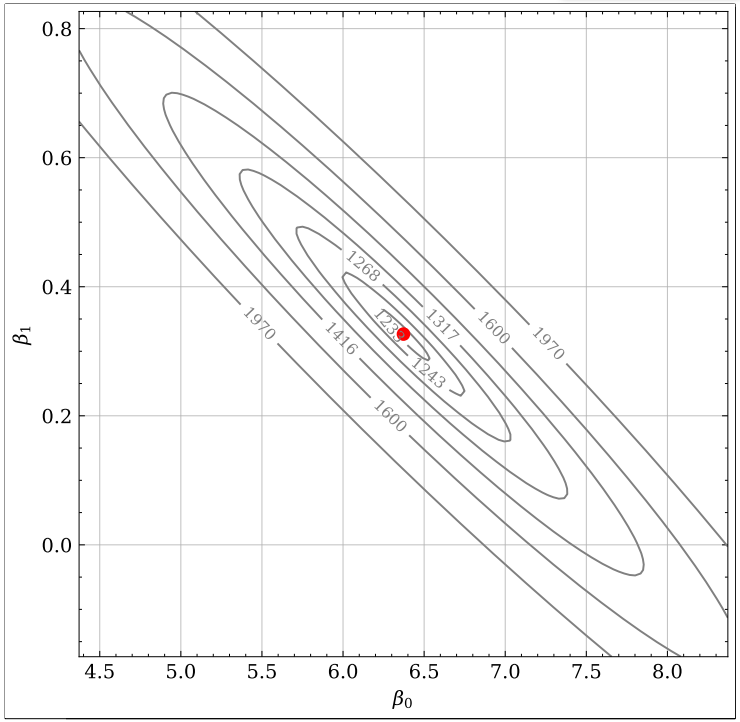


Figure 3.3:rss surface

OLS has a closed form solution.

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= min \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = min \sum_{i=1}^{N} {e_i}^2$$

Residual sum of squares (RSS)

Using some calculus, one can show that the minimizers are:

mean of all x *mean of all y*

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2} = \frac{Cov(X,Y)}{Var(X)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

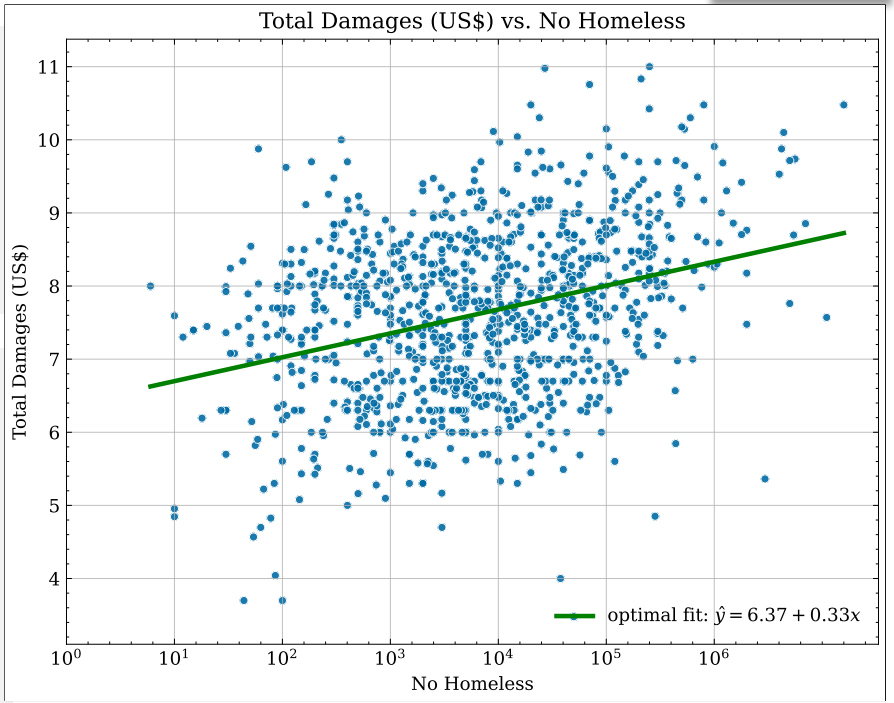


Figure 3.2: ols fit

We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.

Agenda

- Simple linear regression
  - Estimating the Coefficients
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  - Assessing the Accuracy of the Model
- Multiple linear regression
- Extensions to linear regression

Standard errors tell us how much each coefficient would vary across samples.

How good are our estimates $\hat{\beta}_0$ and $\hat{\beta}_1$?

We believe about the world that: $Y = f(X) + \varepsilon$.

… but we cannot observe this entity
… we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$
… from which we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

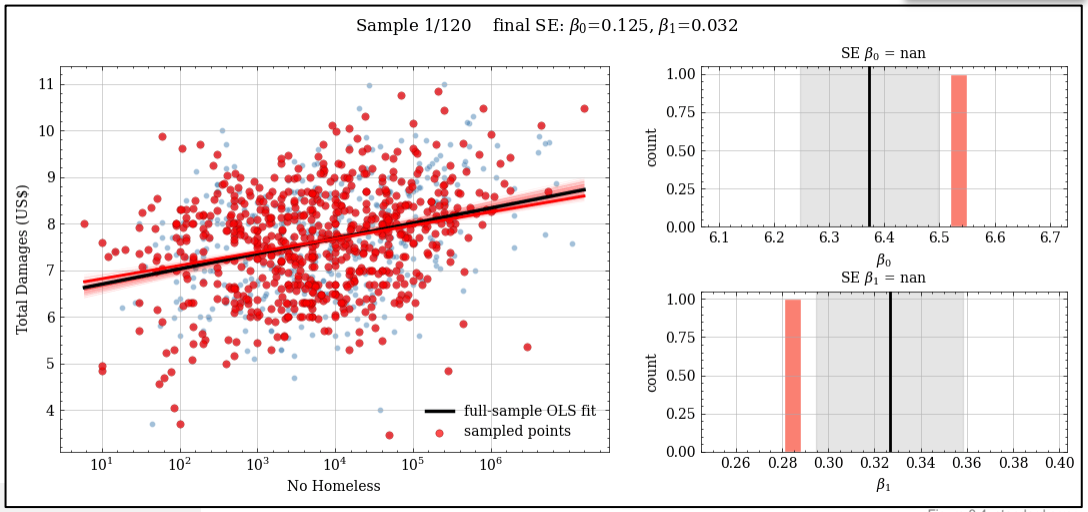


Figure 3.4: standard error

The **standard error** for $\hat{\beta}_0$ and $\hat{\beta}_1$ on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$ are given by:

$$\mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^N (x_i - \bar{x})^2}\right], \quad \mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

, where $\sigma^2 = \mathrm{Var}(\varepsilon)$

How do our observations $\{(x_i, y_i)\}_{i=1}^N$ shape the **standard error**?

Standard errors let us test whether a relationship is real, not noise.

→ **Standard errors** can also be used to perform hypothesis tests on the coefficients.

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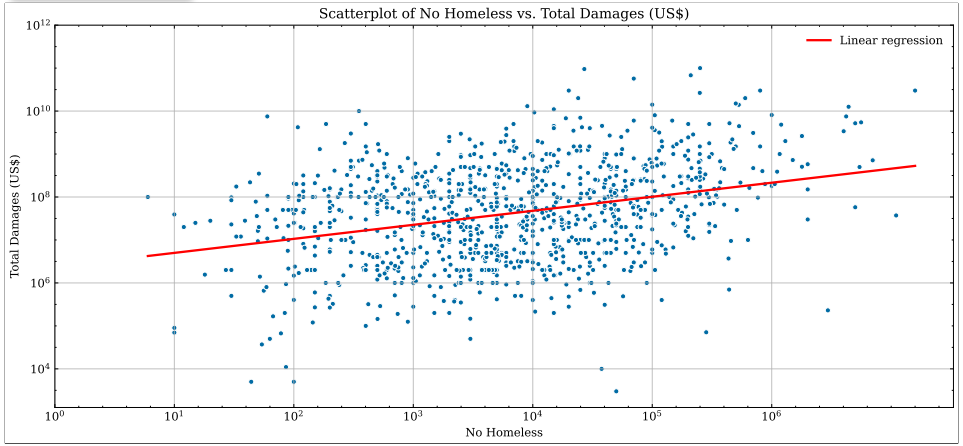


Figure 3.1: line plus noise

Hypothesis testing

H_0: There is no relationship between X and Y versus the alternative hypothesis

H_1: There is some relationship between X and Y

Our question, formalized: does the Homeless feature matter at all?

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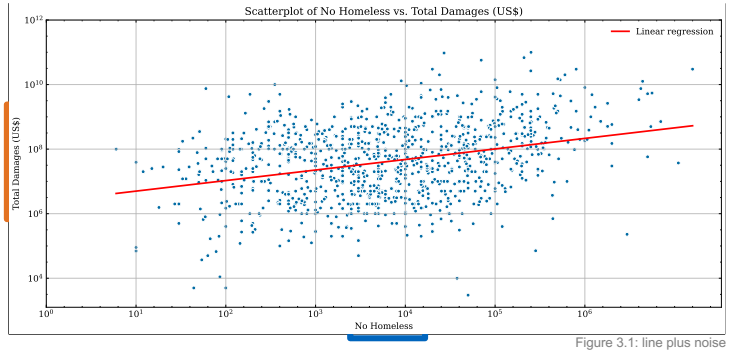


Figure 3.1: line plus noise

Recap that we want to learn a **model**:

$$Total\ Damage\ (US\$) \approx f(Homeless, Total\ Deaths, Injured)$$

Consider the **linear model** and the learned coefficients $\hat{\beta}$ for our case at hand:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \times X_{homeless}$$

Mathematically the hypothesis test for our feature $homeless$ corresponds to:

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

since if $\beta_1 = 0$ then $X_{homeless}$ is not associated with Y.

The t-statistic and p-value quantify the evidence against “no effect”.

$H_0: \beta_1 = 0$
$H_1: \beta_1 \neq 0$

To test the null hypothesis, we compute a **t-statistic,** given by:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

Assuming H_0 is true, we compute the probability of observing a value equal to $|t|$ or larger. This probability is called **p-value**.

→ A small **p-value** indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance.

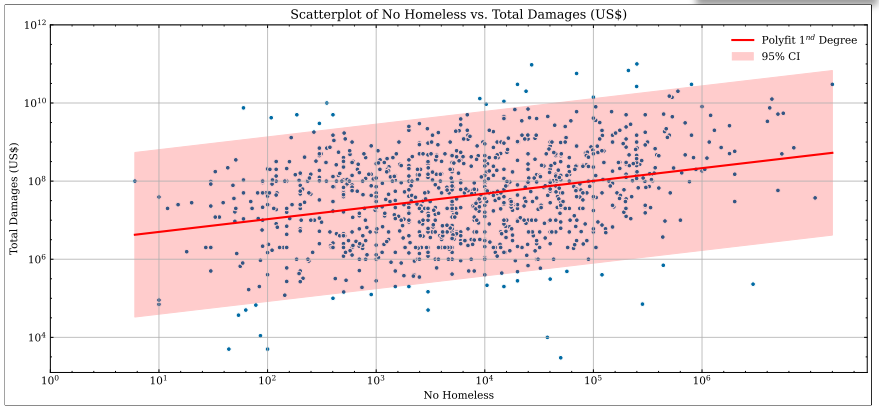


Figure 3.5: confidence band

The **standard error** of an estimator $SE(\hat{\beta}_1)$ reflects how it varies under repeated sampling.

These standard errors can be used to compute confidence intervals. A 95% **confidence interval** is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter:

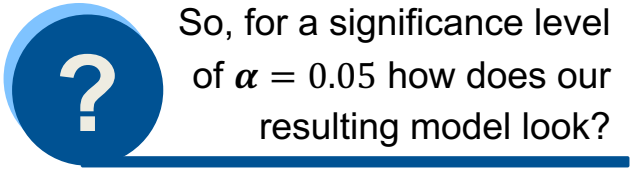
$$[\hat{\beta}_1 - 2{\times}SE(\hat{\beta}_1), \hat{\beta}_1 + 2{\times}SE(\hat{\beta}_1)]$$

Every decision rule risks a false alarm or a missed effect.

		Decision	
		fail to reject H_0	reject H_0
Truth	H_0 true	$\surd$	Type 1 Error
	H_1 true	Type 2 Error	$\surd$

We reject H_0 when the **p-value** is less than the **significance level** $\boldsymbol{\alpha}$. We often set $\boldsymbol{\alpha}$ to be 0.01 or 0.05.

→ This means that, for those cases where H_0 is actually true, we do not want to incorrectly reject it more than 100 x α % of those times.



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OLS Regression Results

Dep. Variable:	Total Damages (US$)	R-squared:	0.103
Model:	OLS	Adj. R-squared:	0.102
Method:	Least Squares	F-statistic:	121.8
Date:	Wed, 17 Jun 2026	Prob (F-statistic):	6.92e-27
Time:	14:06:22	Log-Likelihood:	-1590.3
No. Observations:	1067	AIC:	3185.
Df Residuals:	1065	BIC:	3195.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	6.3726	0.117	54.320	0.000	6.142	6.603
No Homeless	0.3266	0.030	11.035	0.000	0.269	0.385

Omnibus:	4.882	Durbin-Watson:	1.695
Prob(Omnibus):	0.087	Jarque-Bera (JB):	4.813
Skew:	-0.164	Prob(JB):	0.0901
Kurtosis:	3.036	Cond. No.	15.0

Mathematically the hypothesis test corresponds to:

$H_0: \beta = 0$

$H_1: \beta \neq 0$

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RSS alone can't tell us how good the linear assumption is.

Based on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$ *we learned the function:*

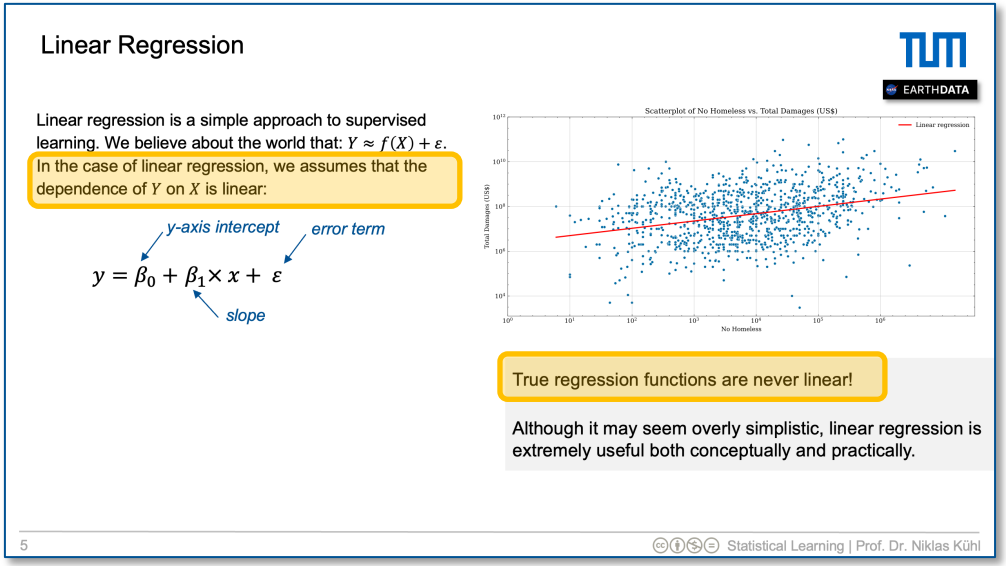
$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

Such that it minimizes the ***Residual sum of squares (RSS):***

$$min \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$



How good is our **linear assumption** in describing our data?



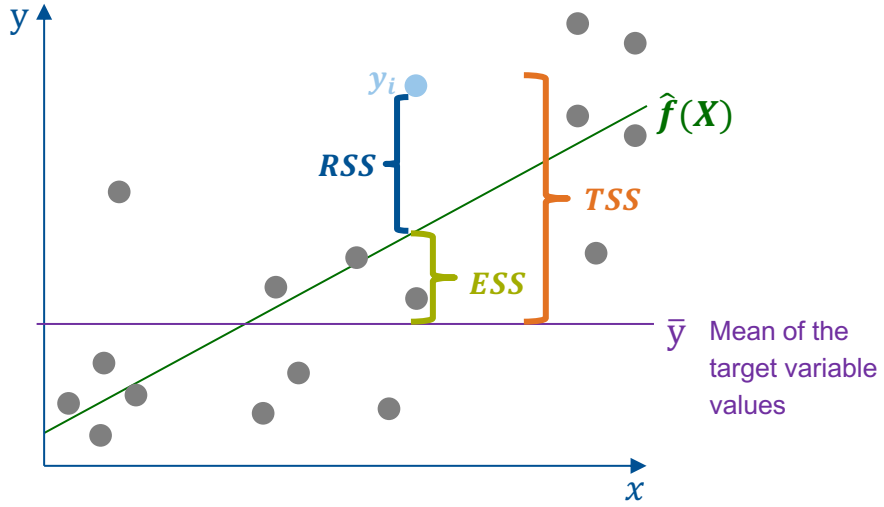
Total variation splits cleanly into explained and residual parts.

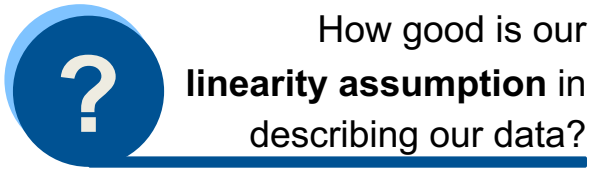
Based on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$ we learned the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

Such that it minimizes the **Residual sum of squares (RSS):**

$$min \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$





How good is our **linearity assumption** in describing our data?

Total sum of squares (TSS) $\sum_{i=1}^{N}(y_i - \bar{y})^2$

= ***Residual sum of squares (RSS)*** $= \sum_{i=1}^{N}(y_i - \hat{y}_i)^2$

+ ***Explained sum of squares (ESS)*** $+ \sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2$

R^2 is the share of variance our line actually explains.



How good is our **linearity assumption** in describing our data?

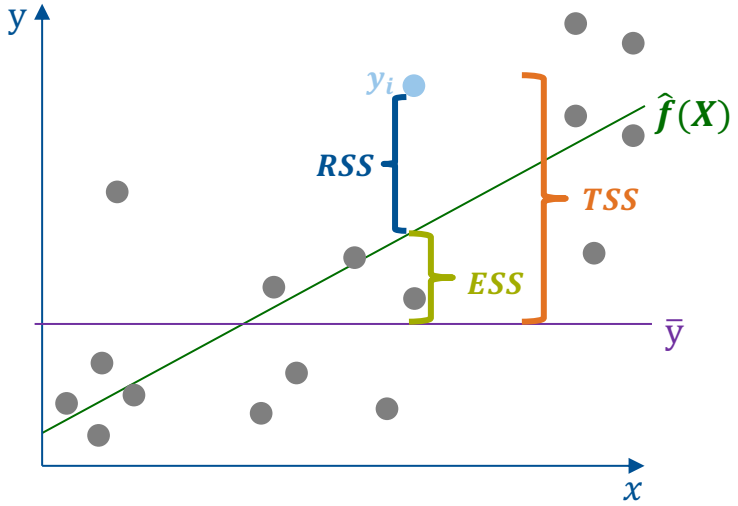
R^2 measures the proportion of the variation in y that is explained by the variation in x:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} = \frac{ESS}{TSS}$$

$$R^2 \in [0, 1]$$

$R^2 = 1$: Perfect match between the line and the datapoints

$R^2 = 0$: no linear relationship between x and y



Total sum of squares (TSS) $\sum_{i=1}^{N}(y_i - \bar{y})^2$

= *Residual sum of squares (RSS)* $= \sum_{i=1}^{N}(y_i - \hat{y}_i)^2$

\+ *Explained sum of squares (ESS)* $+ \sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2$

What do these coefficients actually say about disasters?

Remember our **model**:

$$Total\ Damage\ (US\$) \approx f(Homeless)$$

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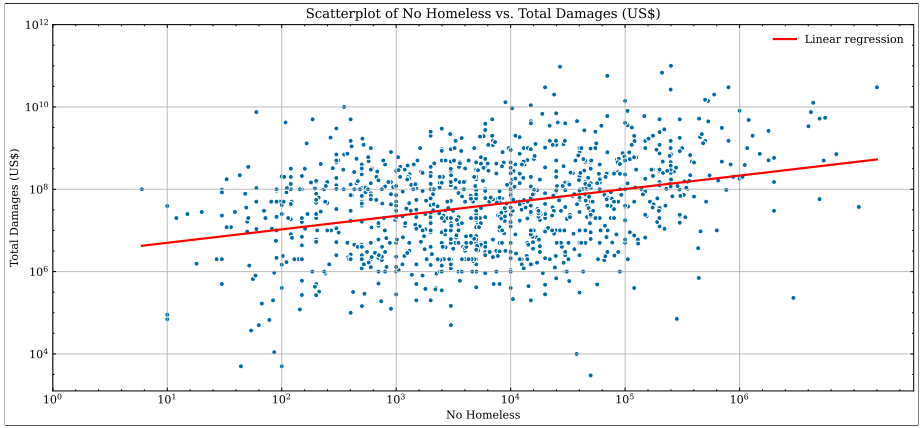


Figure 3.1:line plus noise

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How do we interpret the results?

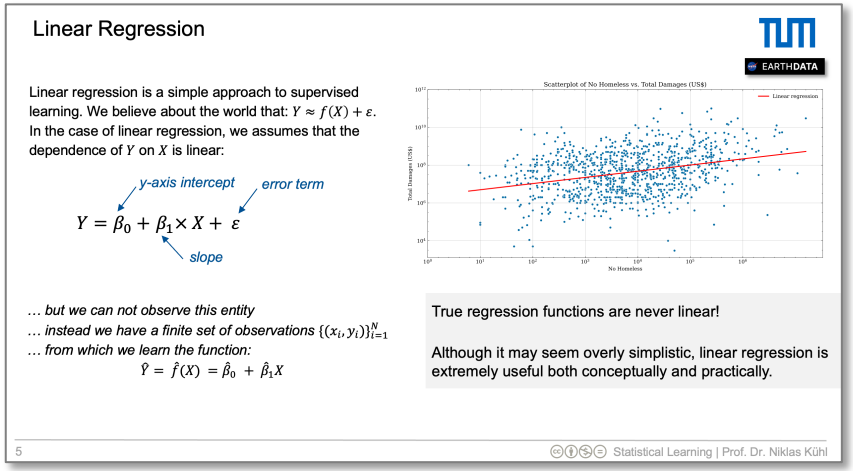


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With many predictors, the prediction is a weighted sum of them all.

What if we now consider: $X = (X_1, X_2, \ldots, X_p)$?



Then our belief about the world is:

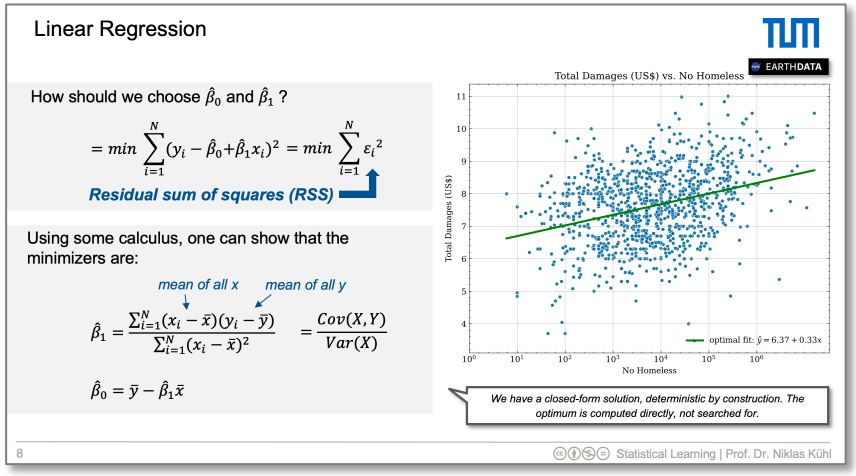
$$Y = \beta_0 + \beta_1 \times X_1 + \beta_2 \times X_2 + \cdots + \beta_p \times X_p + \varepsilon$$

And based on a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$, where x_i is now a p-dimensional vector, we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \cdots + \hat{\beta}_p X_p$$

RSS generalizes unchanged to p predictors.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?



Residual sum of squares (RSS)

$$= \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{\mathrm{i},1} - \hat{\beta}_2 x_{\mathrm{i},2} - \cdots - \hat{\beta}_p x_{\mathrm{i},p})^2$$

Input and target features can be formalized in matrixes.

Feature: an individual measurable property or characteristic of a phenomenon

We let $\boldsymbol{X}$ denote an $n \times p$ matrix whose (i, j) th element is $x_{i,j}$. That is,

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

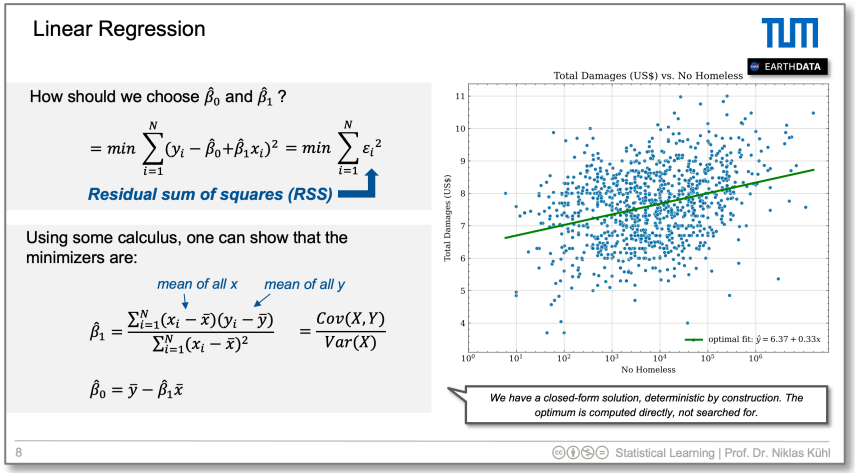
Target: is the variable the model is trying to predict

We use y_i to denote i th observation of the variable on which we wish to make predictions. Hence, we write the set of all n observations in vector form as

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Rewriting RSS in matrix form.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?





Do we still have a closed form optimal solution?

Residual sum of squares (RSS)

$$= \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \mathrm{x}_{\mathrm{i},1} - \hat{\beta}_2 \mathrm{x}_{\mathrm{i},2} - \cdots - \hat{\beta}_p \mathrm{x}_{\mathrm{i},p})^2$$

We formulate in matrix notation $Y = X\beta + \varepsilon$ and get:

$$\sum_{i=1}^{N} (y_i - \mathrm{x}_\mathrm{i}\beta)^2$$

The matrix RSS expands into a term quadratic in β.

We formulate in matrix notation $Y = X\beta + \varepsilon$ and get ***Residual sum of squares (RSS)***

$$= \sum_{i=1}^{N} (y_i - \mathrm{x_i}\beta)^2$$

$$= (\mathrm{y} - \mathrm{X}\beta)'(\mathrm{y} - \mathrm{X}\beta)$$

$$= \mathrm{y}'\mathrm{y} - \mathrm{y}'\mathrm{X}\beta - \beta'\mathrm{X}'\mathrm{y} + \beta'\mathrm{X}'\mathrm{X}\beta$$

$$= \mathrm{y}'\mathrm{y} - 2\beta'\mathrm{X}'\mathrm{y} + \beta'\mathrm{X}'\mathrm{X}\beta$$

Following $\sum_i a_i^2 = a'a$

Expand the product, keep in mind, that transpose reverses the order of the factors $(AB)' = B'A'$

Identify that $\mathrm{y}'\mathrm{X}\beta$ *is of size 1x1 and a scalar equals its own transpose*

Differentiate with respect to β *to identify the minimum* $\frac{\partial RSS}{\partial \beta}$

We differentiate the RSS and set it to zero to identify the optimum.

$$\textbf{RSS} = y'y - 2\beta'X'y + \beta'X'X\beta$$

$$\frac{\partial RSS}{\partial \beta} = \frac{\partial (y'y - 2\beta'X'y + \beta'X'X\beta)}{\partial \beta}$$

$$= \frac{\partial (y'y)}{\partial \beta} - \frac{\partial (2\beta'X'y)}{\partial \beta} + \frac{\partial (\beta'X'X\beta)}{\partial \beta}$$

$$= -2X'y + 2X'X\beta$$

$$\Rightarrow X'X\beta = X'y$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

Differentiate with respect to β *to identify the minimum* $\frac{\partial RSS}{\partial \beta}$

Independent of β → drops

By calculus we know $\frac{\partial (a'b)}{\partial b} = a$

By calculus we know $\frac{\partial (b'Ab)}{\partial b} = 2Ab$ for symmetrical A

Set gradient to zero $\frac{\partial RSS}{\partial \beta} = 0$

If $X'X$ is invertible, multiply both sides on the left by $(X'X)^{-1}$

***Again**: We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.*

Second derivative confirms a minimum.

$$\hat{\beta} = (X'X)^{-1}X'y$$

***Again**: We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.*

? Is it really the min RSS, not the max?

The second derivative (Hessian) is $2X'X$, which is positive semidefinite for any X, so the stationary point is the minimum, not a maximum or saddle.

? How is this solution different to the p=1 case?

It's the same! Set X=[1 x] into $(X'X)^{-1}X'y$ you get $\hat{\beta}_1 = \frac{Cov(X,Y)}{Var(X)}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$

One formula solves least squares for any number of predictors.

What if we now consider: $X = (X_1, X_2, \dots, X_p)$?

Linear Regression

How should we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

$$= min \sum_{i=1}^{N} (y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i)^2 = min \sum_{i=1}^{N} \varepsilon_i{}^2$$

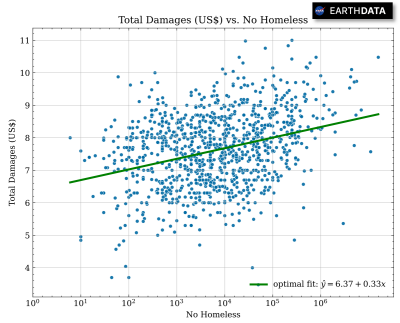
Residual sum of squares (RSS)

Using some calculus, one can show that the minimizers are:

mean of all x — mean of all y

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2} = \frac{Cov(X,Y)}{Var(X)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$



We have a closed-form solution, deterministic by construction. The optimum is computed directly, not searched for.

8 Statistical Learning | Prof. Dr. Niklas Kühl

Residual sum of squares (RSS)

$$= \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \mathrm{x}_{\mathrm{i},1} - \hat{\beta}_2 x_{\mathrm{i},2} - \cdots - \hat{\beta}_p \mathrm{x}_{\mathrm{i},p})^2$$

We formulate in matrix notation $Y = X\beta + \varepsilon$ and get the solution by:

$$\hat{\beta} = (X'X)^{-1}X'y$$

Multiple linear regression



Agenda

- Simple linear regression
- Multiple linear regression
- Extensions to linear regression
  - Dummy variables
  - Modeling feature interaction
  - Coefficient shrinkage methods

We can include qualitative predictors via dummy variables.

Not every predictor is numeric — e.g. we can differentiate events by **Disaster Type** being a wildfire or not. A two-level factor becomes a single 0/1 indicator.

1. Encode the factor as an indicator

$$x_{i,1} = \begin{cases} 1 & \text{if disaster } i \text{ is a Wildfire} \\ 0 & \text{if disaster } i \text{ is any other Disaster Type} \end{cases}$$

Let $x_{i,2}$ be the No. Homeless

2. Put it in the regression

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \beta_2 x_{i,2} + \varepsilon_i & \text{Wildfire} \\ \beta_0 + \beta_2 x_{i,2} + \varepsilon_i & \text{any other} \end{cases}$$

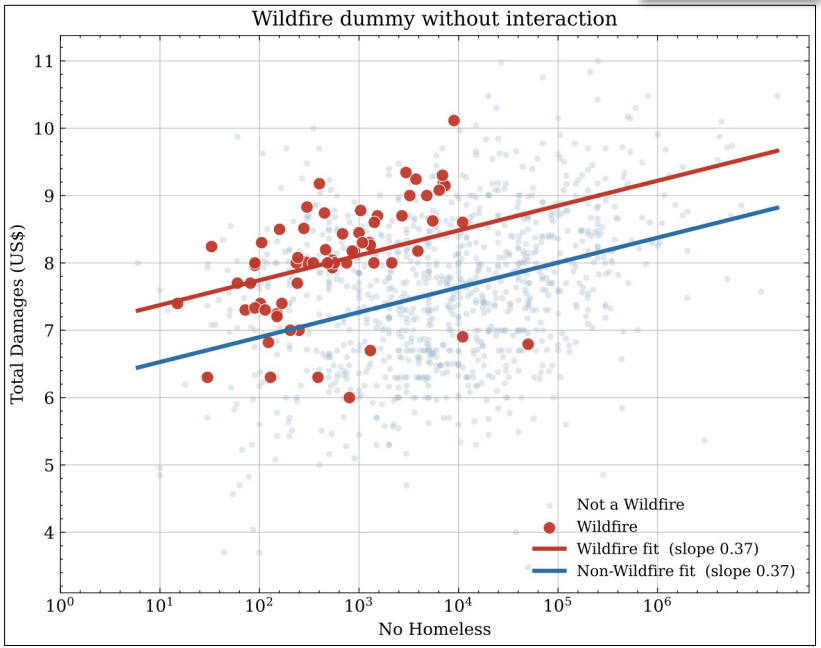


Figure 3.6: dummy variables

How do we expect this regression to look?

What if we believe β_2 to be different for Wildfires?

Agenda

- Simple linear regression
- Multiple linear regression
- Extensions to linear regression
  - Dummy variables
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And we can model interactions along qualitative and quantitative predictors.

Add a product term (Wildfire × log Homeless)

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \boldsymbol{\beta_3\, x_{i,1} x_{i,2}} + \varepsilon_i$$

Split by the two cases

Wildfire ($x_{i,1} = 1$): $y_i = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)\, x_{i,2} + \varepsilon_i$

Otherwise ($x_{i,1} = 0$): $y_i = \beta_0 + \beta_2\, x_{i,2} + \varepsilon_i$

Now the slope on **Homeless** is $\boldsymbol{\beta_2}$ for other disasters but $\boldsymbol{\beta_2} + \boldsymbol{\beta_3}$ for wildfires. $\boldsymbol{\beta_3}$ **is the difference in slope** — non-zero β_3 relaxes the additive assumption, so the lines are no longer parallel.

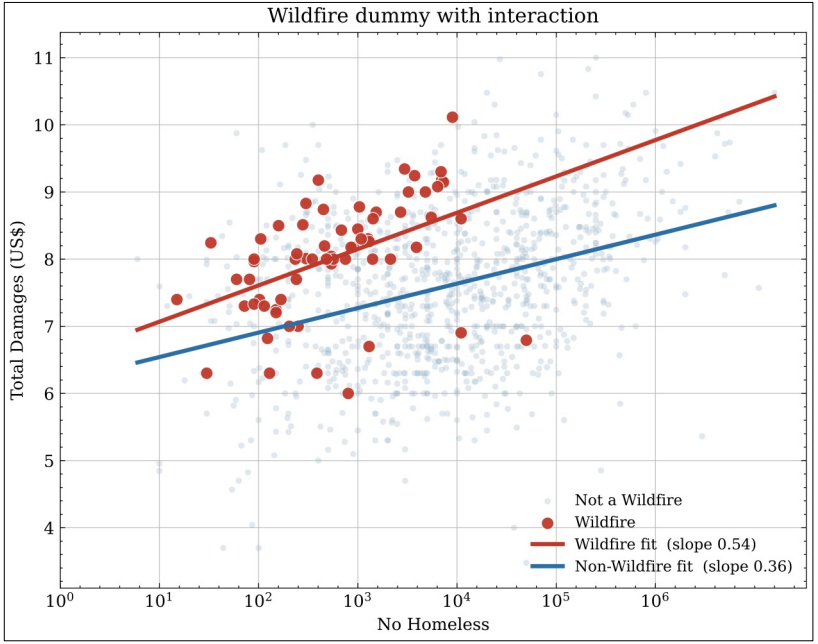


Figure 3.7: interactions

Agenda

- Simple linear regression
- Multiple linear regression
- Extensions to linear regression
  - Dummy variables
  - Modeling feature interaction
  - Coefficient shrinkage methods

To generalize, we deliberately shrink coefficients toward zero.

Recall our objective of minimizing ***Residual sum of squares (RSS)***:

$$min \sum_{i=1}^{N} \left(y_i - \hat{\beta}_0 - \sum_{j=1}^{p} \hat{\beta}_j x_{i,j} \right)^2$$

And its closed form solution:

$$\hat{\beta} = (X'X)^{-1}X'y$$

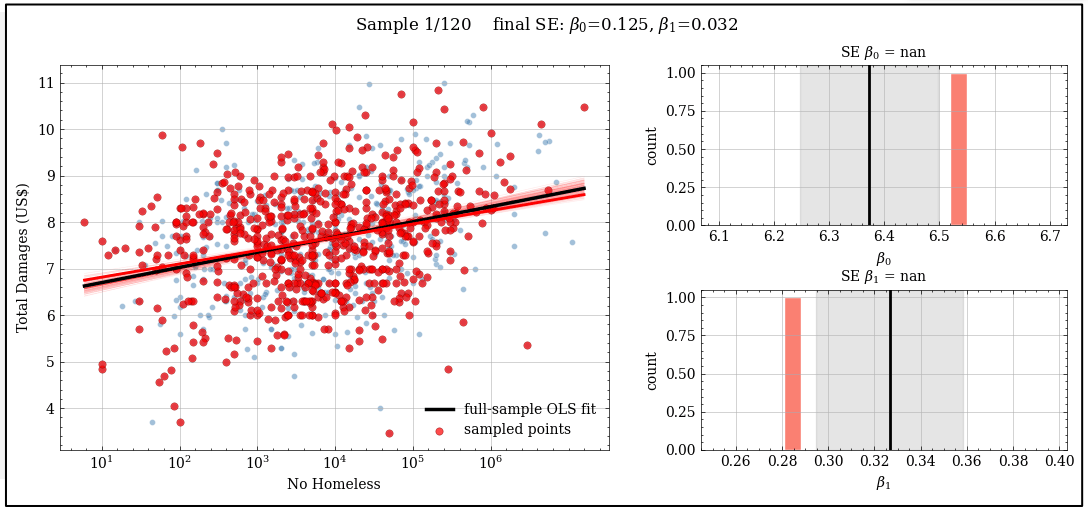


Figure 3.4: standard error

This is great for our finite set of observations $\{(x_i, y_i)\}_{i=1}^{N}$. But eventually we want to generalize to unseen data.

We know that the derived coefficients are subject to variance in the data. And we can account for it by nudging our model towards smaller coefficients by ***shrinkage methods.***

Ridge adds an L2 penalty that shrinks all coefficients smoothly.

We know that the derived coefficients are subject to variance in the data. And we can account for it by nudging our model towards smaller coefficients by ***shrinkage methods.***

We adapt our minimizing objective:

$$min \sum_{i=1}^{N} \left(y_i - \hat{\beta}_0 - \sum_{j=1}^{p} \hat{\beta}_j x_{i,j} \right)^2 + \boldsymbol{\lambda} \sum_{\boldsymbol{j=1}}^{\boldsymbol{p}} \widehat{\boldsymbol{\beta}}_{\boldsymbol{j}}^{\,\boldsymbol{2}}$$

Linear regression with this L-2 regularization term is called ***Ridge regression.***

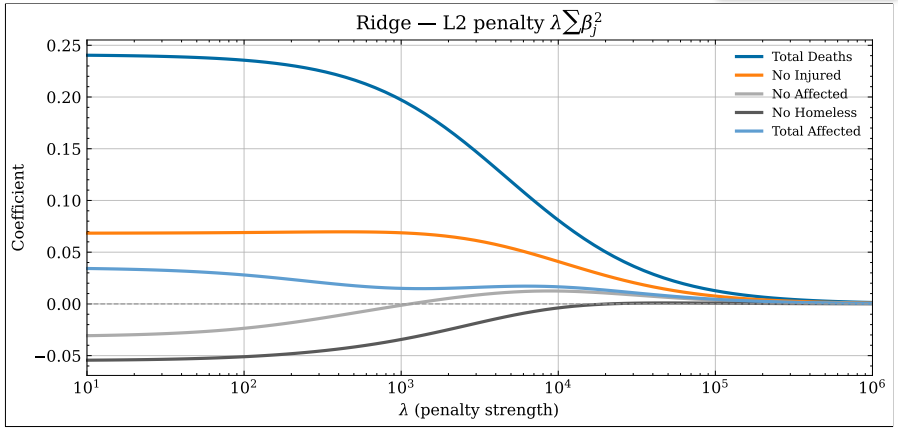


Figure 3.8: ridge

$\boldsymbol{\lambda}$ is a tuning parameter. Selecting a good value for $\boldsymbol{\lambda}$ is critical. As $\boldsymbol{\lambda}$ increases, the flexibility of the ridge regression fit decreases, leading to decreased variance but increased bias.

A few things to keep in mind when applying Ridge regression.

Linear regression with this L-2 regularization term is called ***Ridge regression.***

$$min \;\; RSS + \boldsymbol{\lambda} \sum_{\boldsymbol{j=1}}^{\boldsymbol{p}} \widehat{\boldsymbol{\beta}}_{\boldsymbol{j}}^{\;\boldsymbol{2}}$$

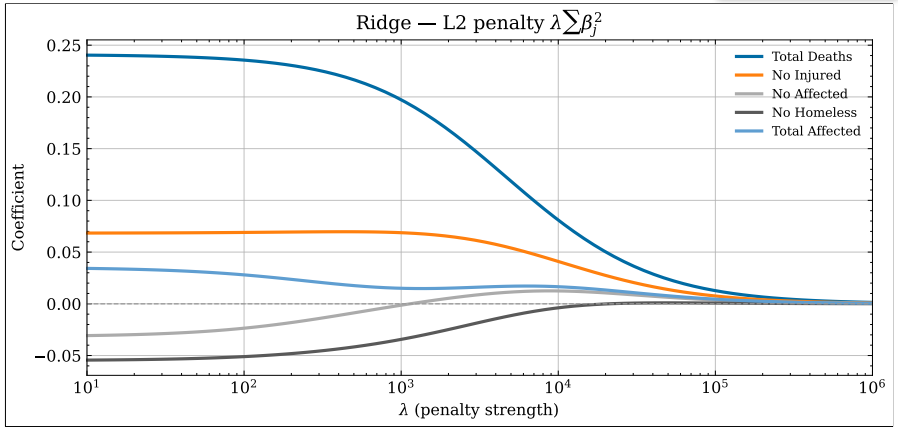


Figure 3.8: ridge

? Why j=1 not j=0?

We do not want to shrink the intercept, which is simply a measure of the mean value of the response variable.

? What becomes critical about our predictors when we apply shrinkage?

Scaling the predictors. Contrast with OLS, which is scale-equivariant: rescale a predictor and its coefficient rescales inversely, leaving predictions, RSS, and significance untouched.

Lasso's L1 penalty can zero coefficients out, doing feature selection.

What if we not only want to shrink coefficients but force our model to select a subset of predictors?

*Linear regression with this L-1 regularization term is called **Lasso regression.***

$$min \;\; RSS + \boldsymbol{\lambda} \sum_{j=1}^{p} \left| \widehat{\boldsymbol{\beta}}_j \right|$$

Why does **L-1 regularization** (***Lasso***) lead to feature selection?

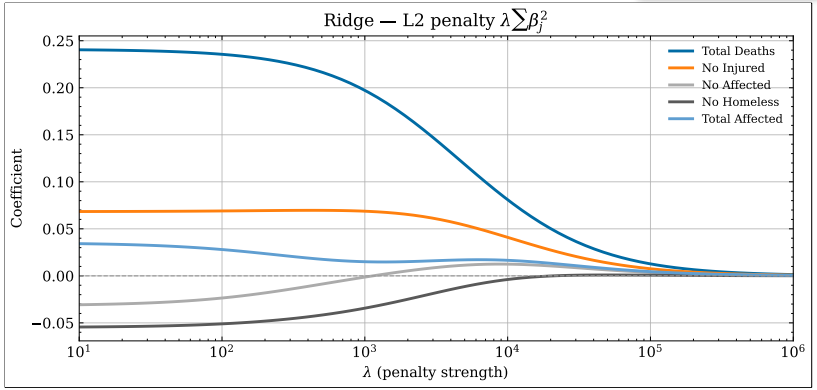


Figure 3.8: ridge

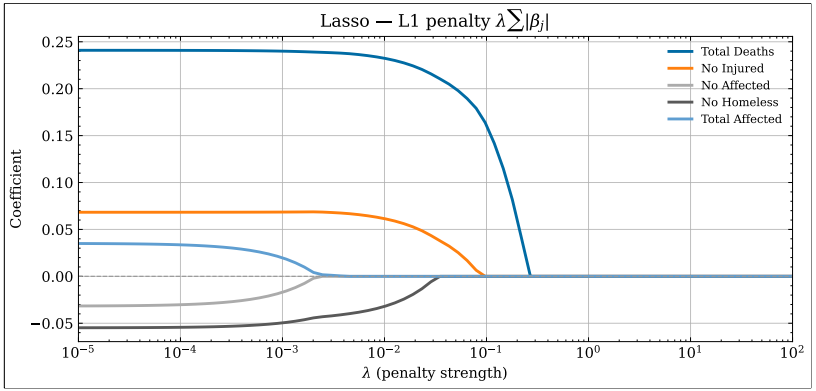


Figure 3.9: lasso

The L1 diamond's corners are why lasso drives coefficients to exactly zero.

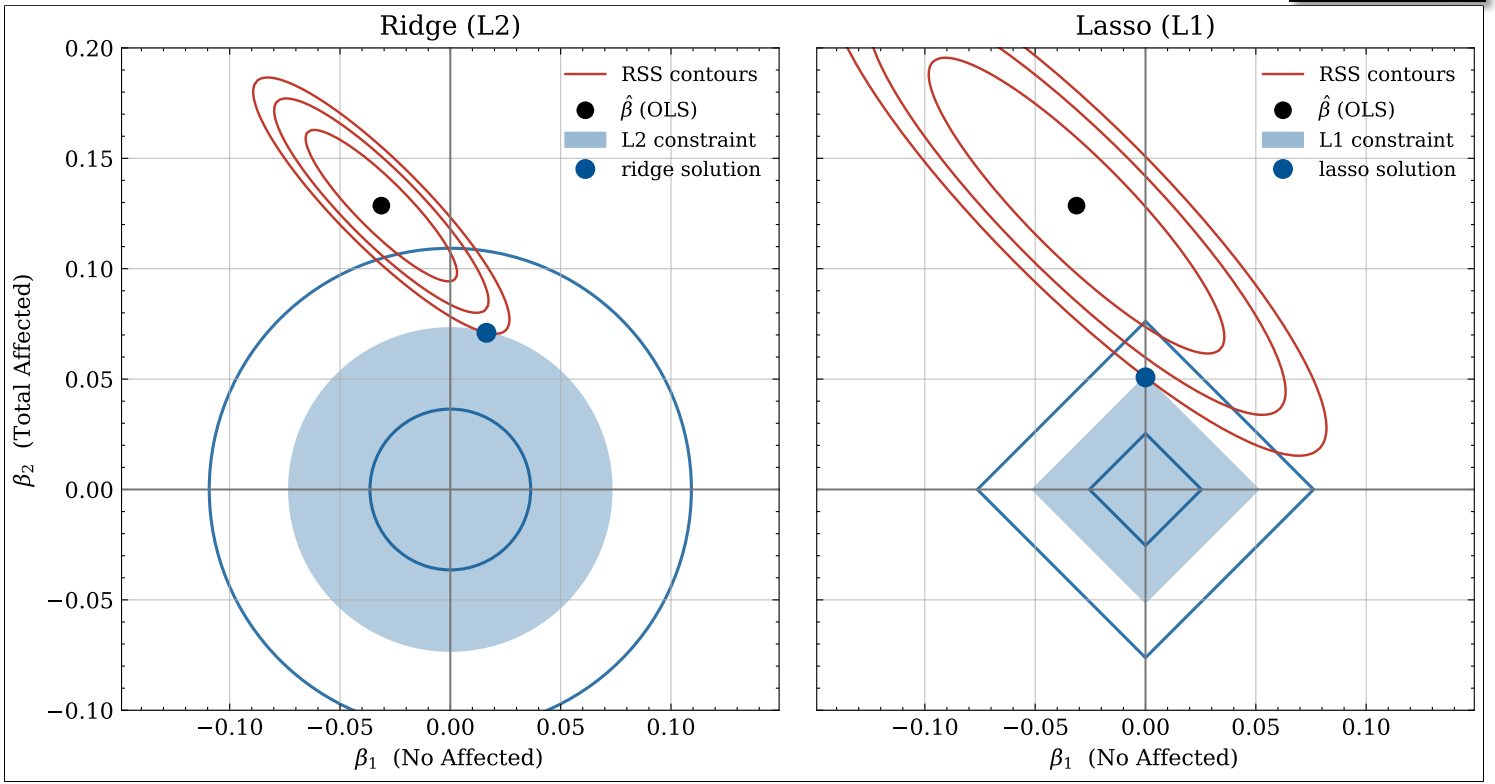
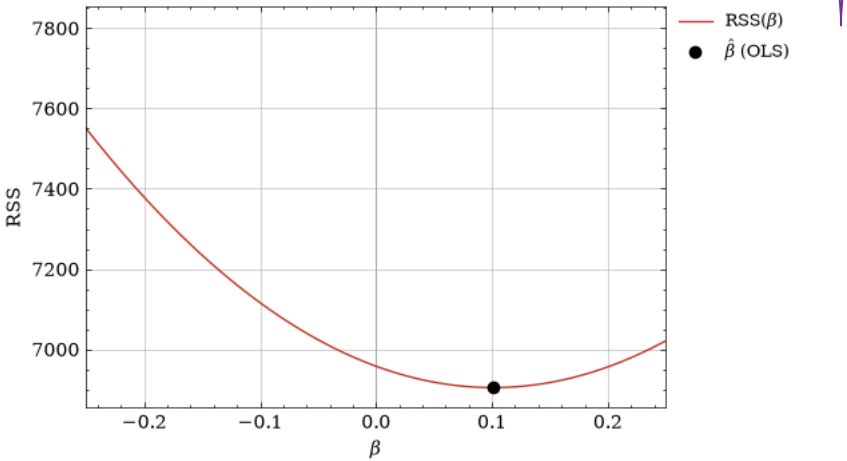


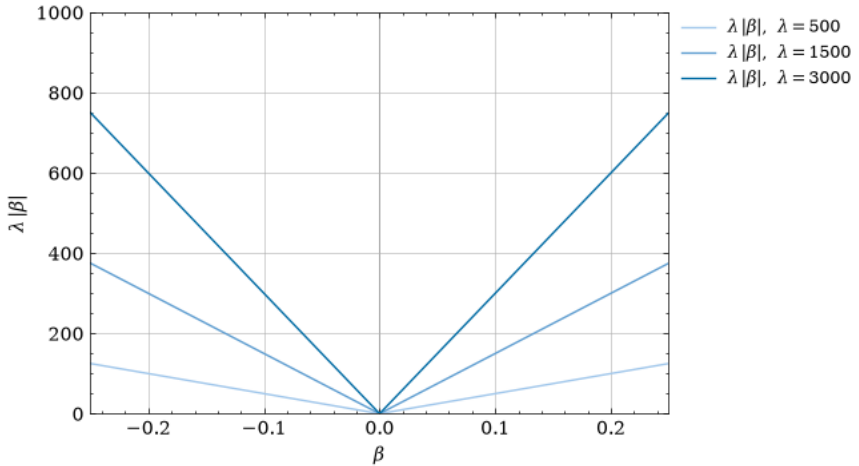
Figure 3.10: constraint geometry

Display both objective terms individually.

Linear regression with this L-1 regularization term is called ***Lasso regression.***

$$min \ \boldsymbol{RSS} + \boldsymbol{\lambda} \sum_{j=1}^{p} |\widehat{\boldsymbol{\beta}}_j|$$

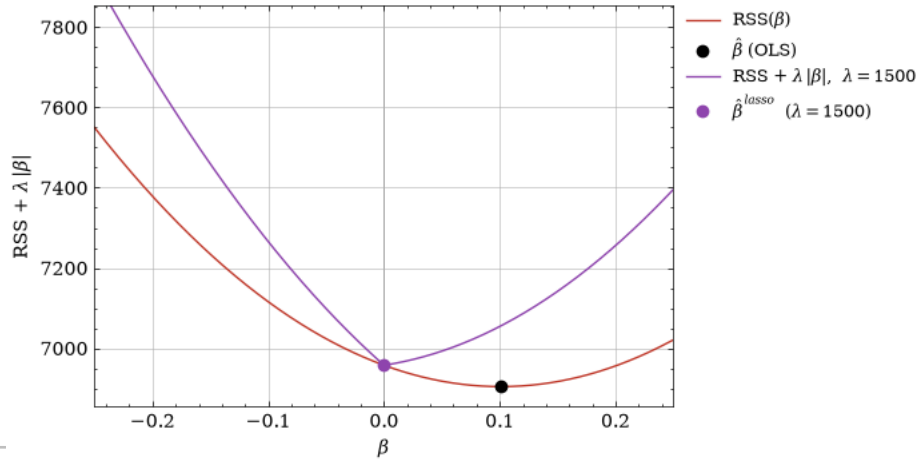




For a given lamba view the resulting objective term.

Linear regression with this L-1 regularization term is called ***Lasso regression.***

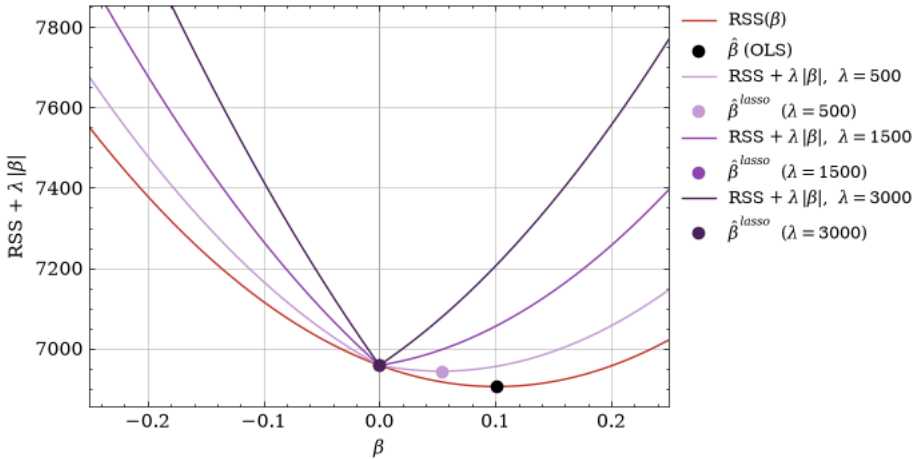
$$min \ \boldsymbol{RSS} + \lambda \sum_{j=1}^{p} |\hat{\beta}_j|$$



For different levels of lambda, we get different optimal solutions.

Linear regression with this L-1 regularization term is called ***Lasso regression.***

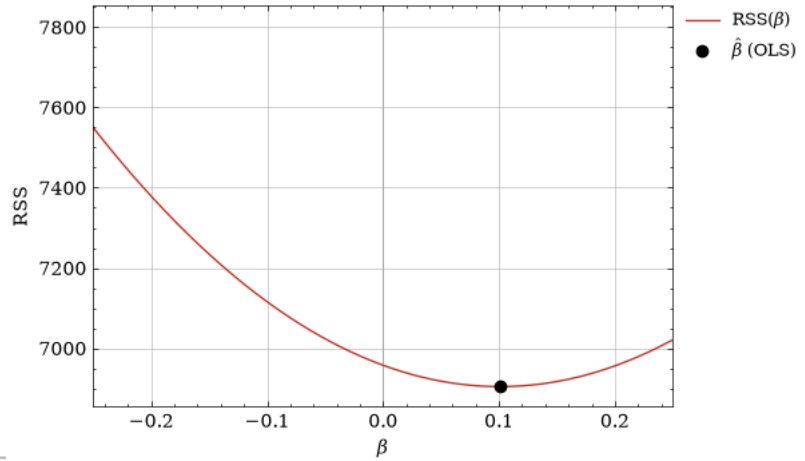
$$min \ \boldsymbol{RSS} + \boldsymbol{\lambda} \sum_{j=1}^{p} |\boldsymbol{\hat{\beta}_j}|$$

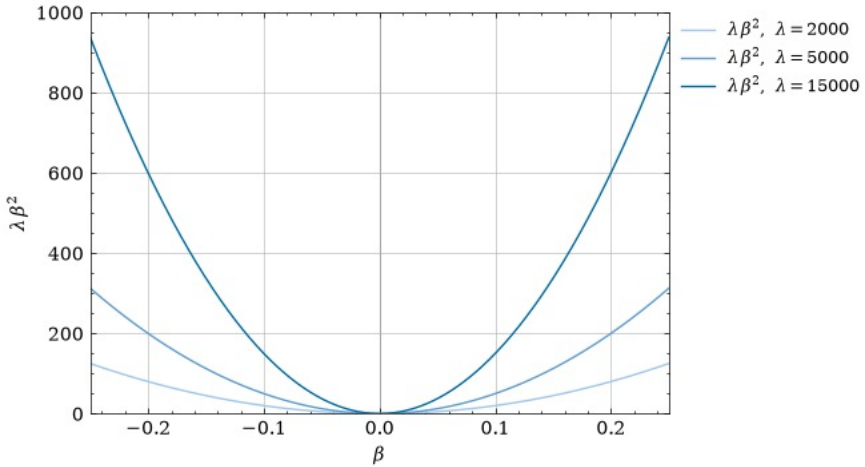


What changes if we apply ridge instead of lasso.

Linear regression with this L-2 regularization term is called ***Ridge regression.***

$$min \ \boldsymbol{RSS} + \boldsymbol{\lambda} \sum_{j=1}^{p} \widehat{\boldsymbol{\beta}}_j^{\,2}$$

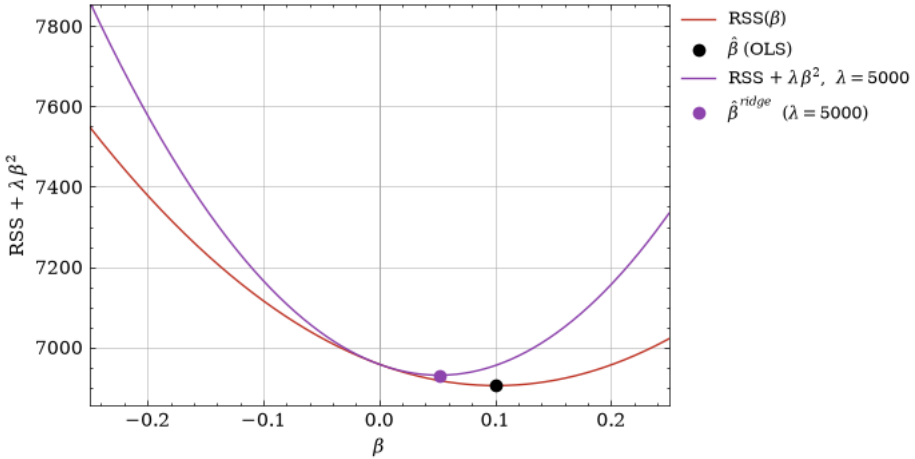




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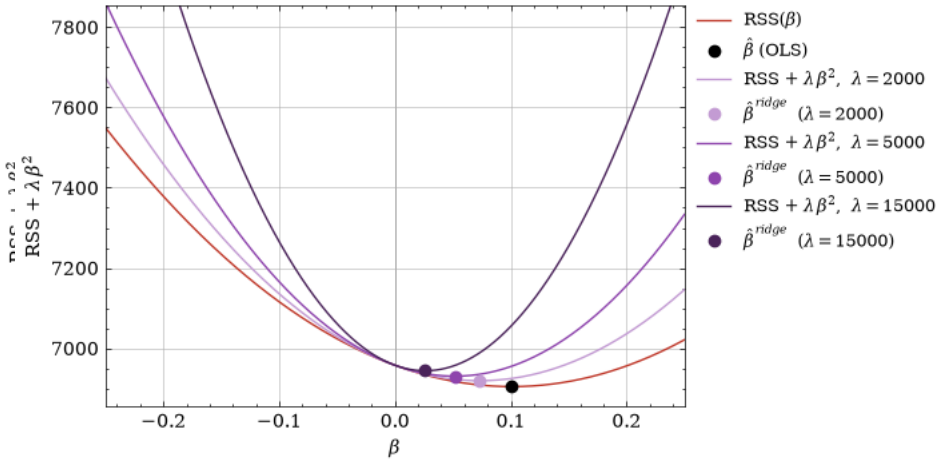
$$\min \; \textbf{RSS} + \lambda \sum_{j=1}^{p} \hat{\beta}_j^{\,2}$$



What changes if we apply ridge instead of lasso.

Linear regression with this L-2 regularization term is called ***Ridge regression.***

$$\min \; \textbf{RSS} \; + \; \lambda \sum_{j=1}^{p} \hat{\beta}_j^{\,2}$$



Then we can move from a single beta to two learned parameters.

